lec 26: Spherical Coordinates

Surface area.

$$\rho = \phi \quad \text{(phi)}$$

$$0 \leq \phi \leq \pi$$

$$\theta = \text{same as before}$$

On a sphere:

$$\rho = a$$

$$\phi \leftrightarrow \text{latitude}$$

$$\theta \leftrightarrow \text{longitude}$$

Polar coordinates in the rz-plane

$$z = \rho \cos \phi$$

$$r = \rho \sin \phi$$

$$x = r \cos \theta = \rho \sin \phi \cos \theta$$

$$y = r \sin \theta = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$\rho = a$$

Sphere of radius a centered at 0.

$$\phi = \frac{\pi}{4}$$

$$z = r$$

$$\phi = \frac{\pi}{2}$$

xy-plane

Triple Integral in Spherical Coordinates

$$dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

$$\Delta V = \Delta\rho \, \Delta\phi \, \Delta\theta$$

Surface Area on Sphere of Radius a

Pieces of circle of radius

$$r = a \sin\phi$$

Length:

$$= a \sin\phi \, \Delta\theta$$

Pieces of circle radius a

$$\text{length} = a\Delta\phi$$

$$\Delta s \approx (a \sin\phi \, \Delta\theta)(a\Delta\phi)$$

$$= a^2 \sin\phi \, \Delta\phi \, \Delta\theta$$

$$ds = a^2 \sin\phi \, d\phi \, d\theta$$

Example

Volume of portion of unit sphere above

$$z = \frac{1}{\sqrt{2}}$$

$$\iiint 1 \, \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

Plane:

$$z = \frac{1}{\sqrt{2}}$$

$$\rho \cos\phi = \frac{1}{\sqrt{2}}$$

$$\rho = \frac{1}{\sqrt{2} \cos\phi}$$

Evaluation

$$\frac{2\pi}{3} - \frac{5\pi}{6\sqrt{2}}$$

Application

Gravitational Attraction

Gravitational force exerted by

$$\Delta m$$

at (x, y, z) on a mass m at origin:

$$|\vec{F}| = \frac{G\,\Delta M\,m}{\ell^2}$$

Direction:

$$\text{dir } \vec{F} = \frac{\langle x, y, z \rangle}{\ell}$$

(unit vector)

$$\vec{F} = \frac{G\,\Delta M}{\ell^2} \langle x, y, z \rangle$$

So integrating,

$$\Delta m = \delta\,\Delta V$$

$$\vec{F} = \iiint \frac{Gm\langle x, y, z \rangle}{\ell^3}\,\delta\,dV$$

Setup: place solid so z-axis is an axis of symmetry. Then

$$\vec{F} = \langle 0, 0, F \rangle$$

$$\iiint \frac{z}{\ell^3} \delta\,dV$$

Use Spherical Coordinates

$$Gm \iiint \frac{\rho \cos\phi}{\ell^3} \delta \, \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

So,

$$F_z = Gm \iiint \cos\phi \sin\phi \, d\rho \, d\phi \, d\theta$$

(z-component)

Example

Newton's theorem:
Gravitational attraction of a spherical planet with uniform density
is equal to that of a point mass with the same total mass at its center.

Lecture 27: Vector Fields in 3D

Vector fields in space.
At every point:

$$\langle P, Q, R \rangle$$

where P, Q, R are functions of x, y, z.

Example

Force fields.

Flux of a Vector Field Through a Surface

$$\vec{F}$$

vector field on a surface in space.

$$\hat{n} = \text{unit normal}$$

There are two choices for $\hat{n}$. Choose a side (orientation).
Flux:

$$\iint_S \vec{F} \cdot \hat{n} \, ds$$

where ds is the surface area element.

Notation

$$d\vec{S} = \hat{n}\, ds$$

Example

Flux of

$$\vec{F} = \langle x, y, z \rangle$$

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \hat{n}\, ds$$

$$\hat{n} = \frac{1}{a} \langle x, y, z \rangle$$

Unit sphere:

$$\sqrt{x^2 + y^2 + z^2} = a$$

$$\vec{F} \cdot \hat{n} = |\vec{F}|\, |\hat{n}|$$

Since

$$\vec{F} \parallel \hat{n}$$

So,

$$\vec{F} \cdot \hat{n} = a$$

$$z = a \cos\phi$$

$$\iint_S \frac{z^2}{a}\, ds = \int_0^{2\pi} \int_0^{\pi} \frac{a^2 \cos^2\phi}{a}\, a^2 \sin\phi\, d\phi\, d\theta$$

$$= 2\pi a^3 \int_0^{\pi} \cos^2\phi \sin\phi\, d\phi$$

$$= 2\pi a^3 \left[-\frac{1}{3} \cos^3\phi \right]_0^{\pi}$$

$$= \frac{4}{3} \pi a^3$$

Conclusion

Use geometry to set up

$$\iint_S \vec{F} \cdot \hat{n}\, ds$$

Horizontal Plane

If

$$S: \; z = a$$

then

$$\hat{n} = \pm\vec{k}$$

and

$$ds = dx\, dy$$

Vertical Plane (yz-plane)

$$x = a$$

$$\hat{n} = \pm\vec{i}$$

$$ds = dy\, dz$$

1. Sphere of Radius a Centered at the Origin

$$\hat{n} = \frac{\langle x, y, z\rangle}{a}$$

$$ds = a^2 \sin\phi\, d\phi\, d\theta$$

2. Cylinder of Radius a Around the z-Axis

$$\hat{n} = \frac{1}{a}\langle x, y, 0\rangle$$

or

$$\hat{n} = \pm\frac{\langle x, y, 0\rangle}{a}$$

$$ds = a\, dz\, d\theta$$

3. Graph $z = f(x, y)$

$$\hat{n}\, ds = \langle -f_x, -f_y, 1 \rangle\, dx\, dy$$

(not unit vector)
To set up bounds on

$$\iint \cdots \, dx\, dy$$

look at the shadow of S in the xy-plane.